Sample of Work 2

Simulation and Pre-conditioned MCMC for Bayesian Logistic and Cauchit Regression

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Code in this report is available at: https://github.com/CloudiaZhang/Data-Sampling-Model-Fitting-and-MCMC-Algorithm (https://github.com/CloudiaZhang/Data-Sampling-Model-Fitting-and-MCMC-Algorithm)

Introduction

This report studies Bayesian binary classification models using logistic and cauchit regression. Firstly, a simulated design matrix and coefficient vector are created, after which binary responses are generated through latent-variable representations. For the cauchit model, latent errors are sampled from the Cauchy distribution directly. For the logistic model, latent errors are generated using rejection sampling with a Cauchy proposal distribution.

Posterior inference is then carried out for four fitted models: logistic and cauchit regression under a standard Gaussian prior and a Unit Information Prior. To improve on a basic random-walk Metropolis algorithm, a pre-conditioned Gaussian proposal is used together with longer MCMC runs, with the aim of improving mixing and effective sample size. The fitted models are assessed using trace plots, autocorrelation, effective sample size summaries, posterior recovery of the true coefficients, and predictive performance measured by the Brier score.

Data Generation

A simulated design matrix with $n = 150$ observations and $d = 10$ predictors is first constructed. The first column is set equal to 1 to represent the intercept term, while the remaining columns are generated from Gaussian random variables and transformed so that several predictors are correlated and have different scales. The true regression coefficient vector is then defined as a sequence of 10 equally spaced values from -2 to 2.

In the next step, n observations Y_i from a cauchit regression model are generated using the latent variable representation given above. The linear predictor is first calculated, and the latent variable $Y_i{}^*$ is defined by

$$Y_i{}^* = x_i^T \beta + \epsilon_i.$$

Since in the cauchit regression model the error term follows a Cauchy distribution, which is equivalent to a t distribution with 1 degree of freedom, it can be simulated using the function `rt` . The latent variable $Y_i{}^*$ is then generated, and the binary observations Y_i are obtained using $Y_i = I(Y_i{}^* > 0)$. The result is a set of simulated realizations of Y_i, where Y_i takes the value 1 if $Y_i{}^* > 0$ and 0 otherwise.

```
## Binary outcomes generated from the cauchit model:
```

```
##   [1] 1 0 0 0 1 0 0 0 0 1 1 0 1 0 1 1 0 0 0 1 0 0 0 0 0 0 0 1 0 1 1 1 0 0 1 0 0
##  [38] 0 1 0 1 1 0 0 1 0 1 1 1 0 1 0 1 1 0 1 0 0 0 0 1 1 1 1 1 1 1 1 0 1 0 0 0 0
##  [75] 1 0 1 0 1 0 1 1 1 0 1 1 1 0 0 1 1 1 0 1 1 1 1 1 1 0 1 0 0 0 1 0 0 0 1 0 1
## [112] 0 0 0 1 0 0 0 0 0 0 1 0 0 0 0 1 0 1 0 0 0 1 0 1 1 1 0 1 1 0 0 1 1 1 0 0 1
## [149] 0 1
```

Rejection Sampling Algorithm

Then n ϵi values are simulated from the logistic distribution using rejection sampling algorithm, with Cauchy distribution being the proposed candidate distribution. Firstly, the density of target distribution π(x) is defined as the logistic distribution with density

$$f_L(x) = \frac{e^{-x}}{(1+e^{-x})^2}$$

The density of candidate distribution q(x) is defined as the Cauchy distribution with density

$$f_C(x) = \frac{1}{\pi(1+x^2)}$$

An appropriate value of M should be defined, which is the maximum of the ratio of the density of target distribution over density of candidate distribution. To make M finite, the candidate density q(x) must have tails that are at least as heavy as those of targeted density π(x). In this case, q(x) is Cauchy distribution which indeed has heavier tails. To find an appropriate value of M, the plot of the ratio of two densities is drawn. By adjusting the x-axis to be within (-10,10), we can see a clear pattern of the plot with the maximum value appears to be around 1.7. By evaluating the values of the ratio at x values between -10 to 10, we can find the maximum value of the ratio being around 1.65, which is the appropriate value of M to be used in the algorithm.

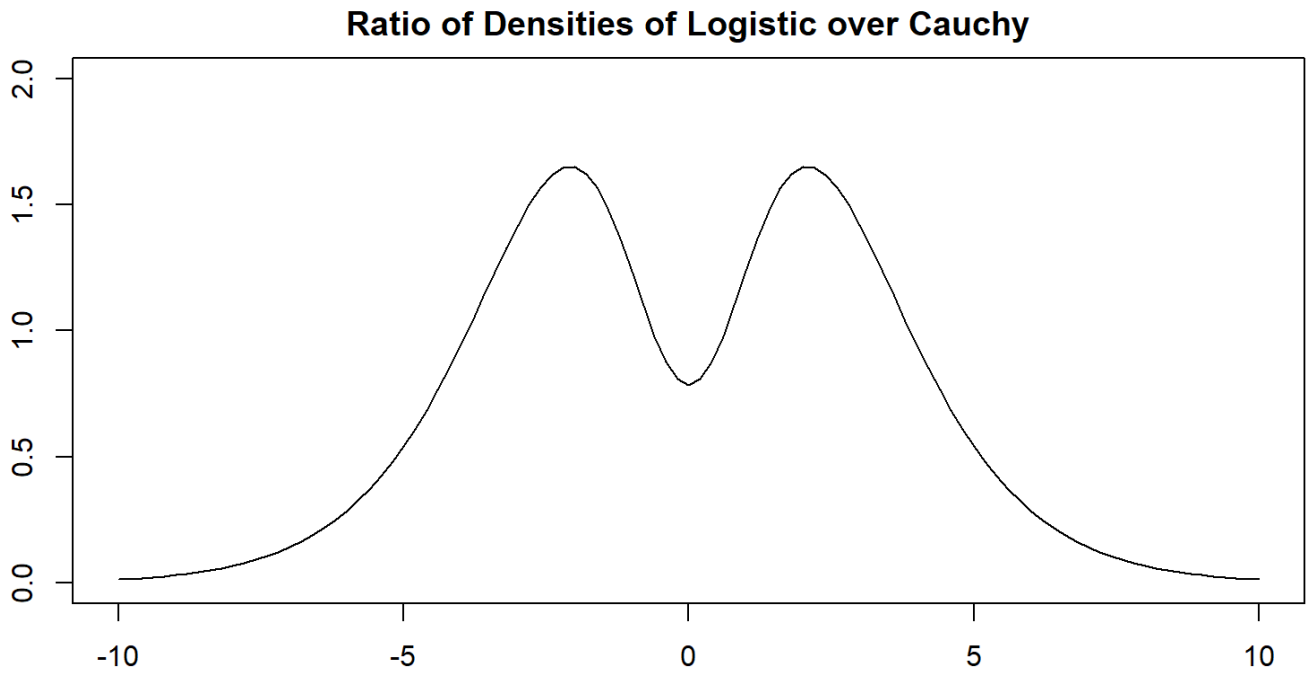

Figure 1: Ratio of Densities

Then the rejection sampling algorithm is applied. A candidate value is drawn from the Cauchy distribution, and u is drawn from a Uniform distribution $U(0,1)$. If u is smaller than the acceptance probability

$$\frac{f_L(x)}{M \cdot f_C(x)},$$

the candidate is accepted. This process is repeated until $n = 150$ accepted samples have been obtained. These accepted values are then used to construct the latent variable for the logistic regression model and generate the binary outcomes Y_i. The resulting values of Y_i for both models are shown in Appendix 1.

The efficiency of the algorithm is influenced by the choice of M, since a smaller value of M generally leads to a higher acceptance rate. In this case, M is not large, so the empirical acceptance rate remains reasonably high, meaning that the required number of accepted samples can be obtained without an excessive number of candidate draws. Therefore, the use of the Cauchy distribution as the candidate distribution provides a reasonably efficient rejection sampling scheme for generating logistic latent errors.

```
## Binary outcomes generated from the logistic model:
```

```
##   [1] 1 0 0 1 1 0 0 0 0 1 1 0 1 0 1 1 0 0 0 1 0 0 0 0 0 0 0 1 1 1 1 1 0 1 1 0 0
##  [38] 0 1 0 1 1 0 0 0 0 1 1 1 0 1 0 0 1 0 1 0 0 0 0 1 1 1 1 1 1 1 1 0 1 0 0 0 0
##  [75] 1 0 1 0 1 0 1 1 1 0 0 1 1 0 0 1 1 1 0 1 1 1 1 1 1 0 1 0 0 0 1 0 0 0 1 0 1
## [112] 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0 1 0 1 1 0 0 1 1 0 0 1 1 1 0 0 1
## [149] 0 1
```

```
## Acceptance rate: 0.6302521
```

```
## Estimated M: 1.651978
```

Metropolis–Hastings Algorithm

The Metropolis–Hastings algorithm is used to fit both the logistic and cauchit regression models to the simulated datasets. Firstly, a standard Gaussian prior is chosen, where each coefficient β_j follows $N(0,1)$. Then the prior is changed to the Unit Information Prior, given by

$$\beta \sim N_d\left(0,\, n(X^TX)^{-1}\right).$$

For the logistic regression model, the probability of success is

$$P(Y_i = 1 \mid X = x_i) = \frac{1}{1+e^{-x_i^T\beta}}.$$

For the cauchit regression model, the probability of success is

$$P(Y_i = 1 \mid X = x_i) = \frac{1}{\pi}\arctan(x_i^T\beta) + \frac{1}{2}.$$

In both models, the binary response is fitted using a Bernoulli likelihood, and the log-likelihood is combined with the log-prior to form the log-posterior.

After that, a pre-conditioned Random-Walk Metropolis algorithm is applied. Compared with the most basic random-walk proposal, this proposal uses a covariance matrix based on $(X^TX)^{-1}$, so that some of the scale and dependence structure in the predictors can be taken into account. This is done to improve the mixing of the chain and to increase the effective sample size. A small diagonal adjustment is also added in the code to make the matrix numerically stable.

Using the Gaussian prior and the Unit Information Prior, four fitted models are obtained in total: logistic regression with Gaussian prior, cauchit regression with Gaussian prior, logistic regression with Unit Information Prior, and cauchit regression with Unit Information Prior. In each case, the chain is started at the true coefficient vector and run for 50,000 iterations. The posterior samples and acceptance rates are stored for later evaluation and comparison. The first few results and the acceptance rates are displayed in Appendix 2.

```
## Acceptance rate (logistic regression with Gaussian prior): 0.45706
```

```
## Acceptance rate (cauchit regression with Gaussian prior): 0.45602
```

```
## Acceptance rate (logistic regression with Unit Information prior): 0.73444
```

```
## Acceptance rate (cauchit regression with Unit Information prior): 0.72438
```

Evaluation

Firstly, the trace plots of the four fitted models are generated. The trace plots show how the parameter samples move over the iterations of the MCMC process. In this case, the plots are drawn after removing the first 10,000 iterations as burn-in. Overall, the four trace plots fluctuate around a central value without any obvious trend, which suggests that the chains are reasonably stable

after burn-in. The two logistic models appear to move slightly more smoothly across the parameter space, while the cauchit models show slightly greater local variation, but there is no clear sign that any of the chains has failed to mix after burn-in.

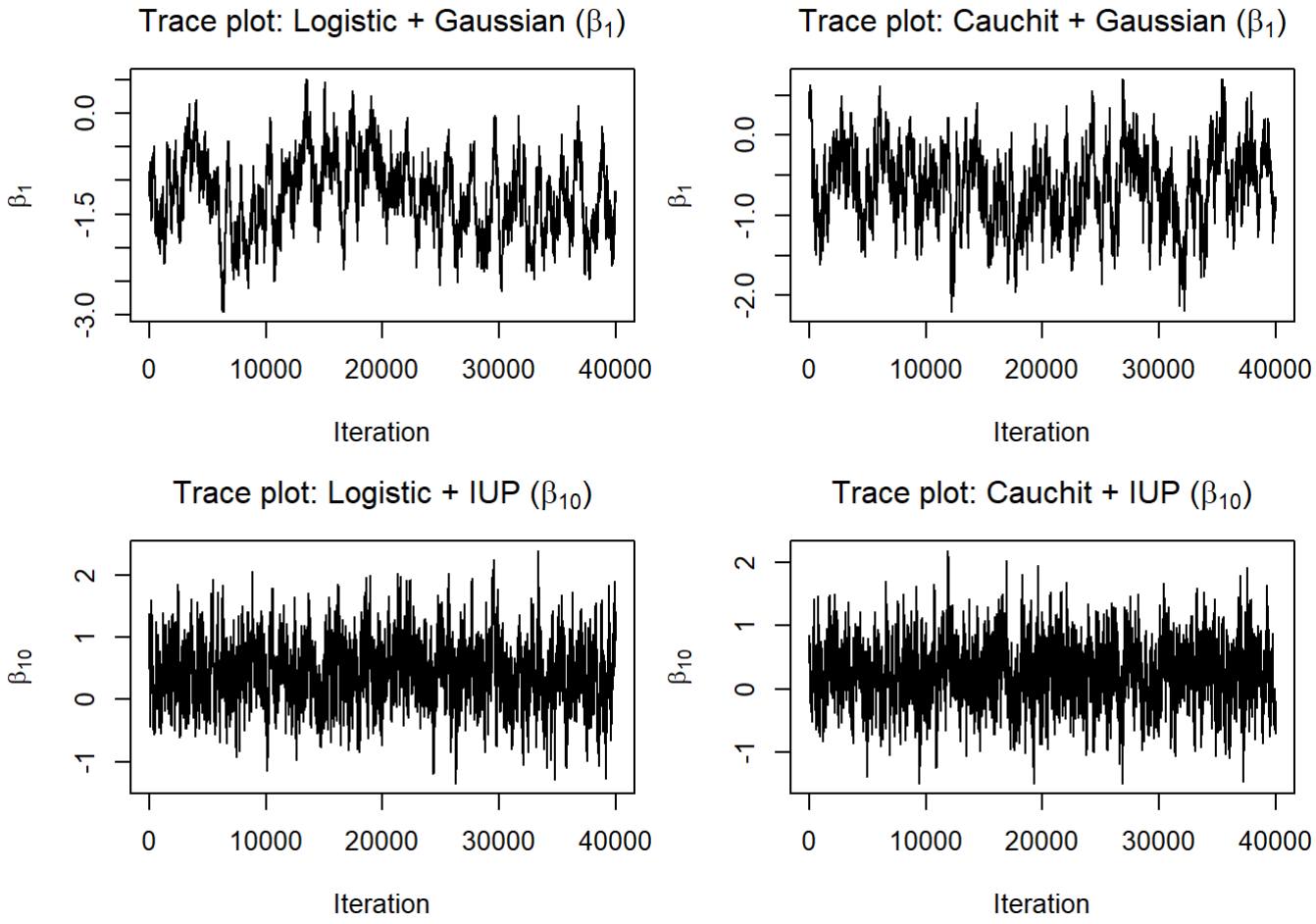

Figure 2: Trace plots

Secondly, the histograms of effective sample sizes for each MCMC sampling are examined. The results show quite different patterns across the four fitted models. For the two models with Gaussian prior, the effective sample sizes are more uneven, with some parameters having relatively small ESS values while a few have much larger values. This suggests that the mixing is not equally good across all parameters, and some coefficients may still be affected by stronger autocorrelation.

In contrast, the models fitted with the Unit Information Prior show more concentrated effective sample sizes, with most of the values lying in the range of around 300 to 450. This suggests a more stable level of sampling efficiency across the parameters. Comparing the logistic and cauchit models under the Unit Information Prior, the cauchit model has a slightly higher median ESS, although the difference is not very large. Overall, the Unit Information Prior appears to give more consistent effective sample sizes than the Gaussian prior in this case.

```
## ESS summary table:
```

```
##                        min median     max
## Logistic + Gaussian  63.57 235.27 5553.77
## Cauchit + Gaussian   50.69 170.40 5479.19
## Logistic + IUP      302.99 365.70  389.73
## Cauchit + IUP       343.47 384.56  451.73
```

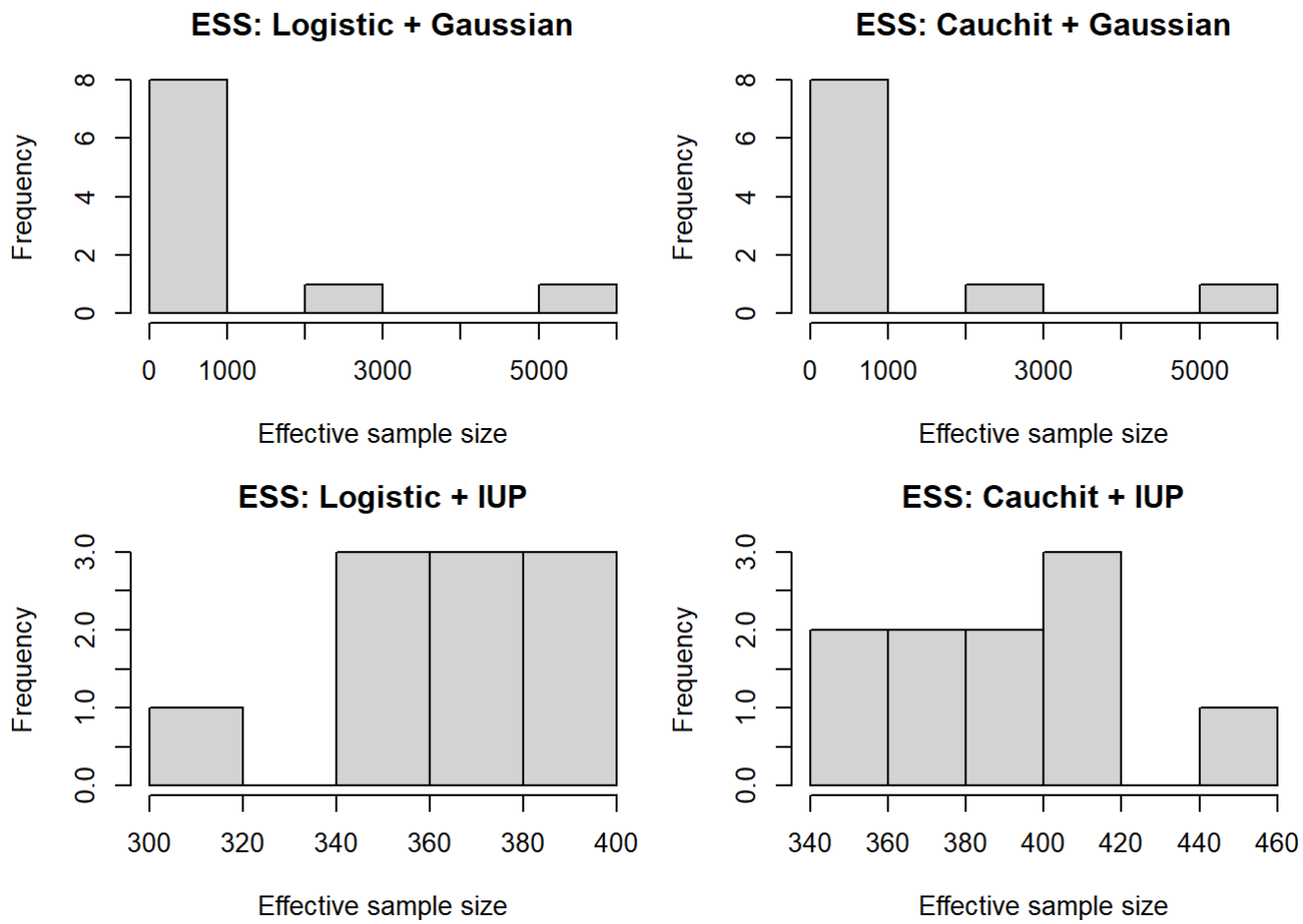

Figure 3: Histograms of Effective Sample Size

Thirdly, ACF values are calculated for each case to assess the autocorrelation in the MCMC samples. The ACF values remain high for all four models, which suggests that autocorrelation is still present in the MCMC samples. However, the two models fitted with the Unit Information Prior show a clearer decay in ACF than the two models with Gaussian prior, indicating an improvement in mixing. Between the logistic and cauchit models, the difference is not very large, although the logistic model with Unit Information Prior appears to decay slightly faster. Overall, autocorrelation remains an issue, but the Unit Information Prior gives better ACF behaviour in this case.

Fourthly, the posterior mean estimates are also compared with the true coefficient values. In these plots, points closer to the dashed 45-degree line indicate better recovery of the true coefficients. Overall, the fitted models capture the general direction of the coefficients, but the agreement with the true values is not very close for all parameters. Among the four models, the logistic model with Gaussian prior appears to follow the overall trend more closely, while the other models show larger deviations for several coefficients. This suggests that the fitted models are able to recover part of the underlying pattern, although the estimation accuracy is still limited for some of the coefficients.

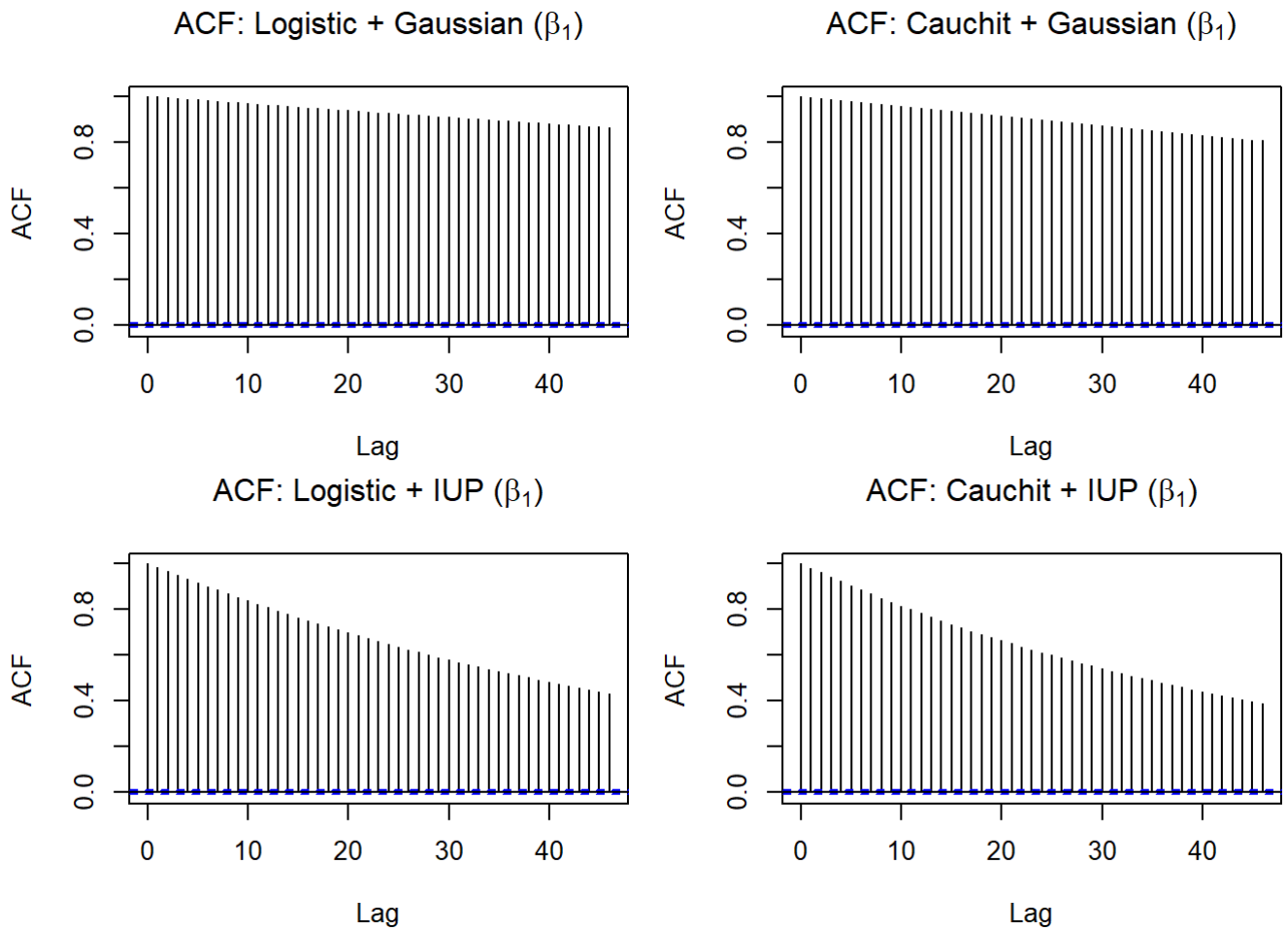

Figure 4: ACF for parameter samples

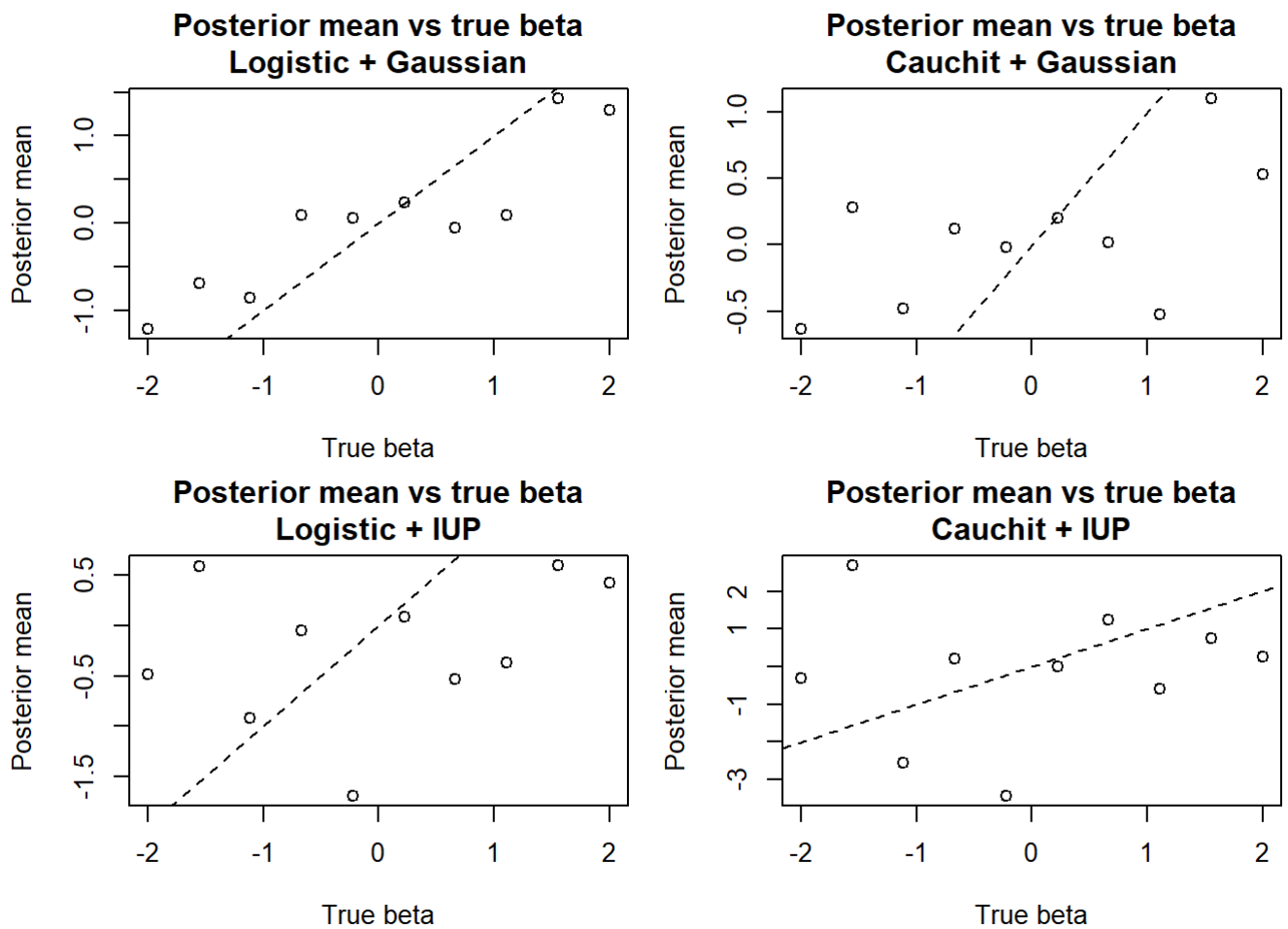

Figure 5: Posterior mean estimates compared with the true coefficient values

Finally, the Brier score is used to assess the quality of predictions. The results show some clear differences across the four fitted models. Among them, logistic regression with Gaussian prior has the lowest Brier score, which indicates the best predictive performance in this case. Logistic regression with Unit Information Prior also performs reasonably well, although its Brier score is higher than that of logistic regression with Gaussian prior. In contrast, the two cauchit regression models have larger Brier scores, especially the model with Unit Information Prior, which suggests weaker predictive performance. Therefore, unlike the earlier diagnostics where the Unit Information Prior gave more consistent effective sample sizes, the Brier scores suggest that logistic regression with Gaussian prior provides the strongest prediction among the four models.

The acceptance rates also differ across the four models. The two models with Gaussian prior have acceptance rates of around 0.46, while the two models with Unit Information Prior have much higher acceptance rates, above 0.72. This suggests that the chains under the Unit Information Prior are making smaller or easier accepted moves more often, although a higher acceptance rate does not necessarily imply better overall predictive performance.

```
##
## Brier scores:
```

```
##                        [,1]
## Logistic + Gaussian 0.007856
## Cauchit + Gaussian  0.044638
## Logistic + IUP      0.031939
## Cauchit + IUP       0.055306
```

```
##
## Acceptance rates:
```

```
##                      [,1]
## Logistic + Gaussian 0.4571
## Cauchit + Gaussian  0.4560
## Logistic + IUP      0.7344
## Cauchit + IUP       0.7244
```

Conclusion

In summary, the analysis of the MCMC outputs for logistic and cauchit regression models with two different priors shows some clear differences across the four fitted models. From the trace plots, all four chains appear reasonably stable after burn-in, although the logistic models seem slightly more well-behaved overall. The effective sample size results suggest that the Unit Information Prior gives more consistent sampling efficiency across parameters, while the Gaussian prior leads to more uneven ESS values. However, the ACF plots still show that autocorrelation remains present in all four models.

The posterior mean plots show that the general pattern of the true coefficients is captured, although the fitted values do not match the true values closely for all parameters. Among the four cases, the logistic model with Gaussian prior appears to follow the overall trend slightly better. The Brier scores show clearer differences in predictive performance, with logistic regression under Gaussian prior giving the lowest score and therefore the best prediction in this case. In contrast, the two cauchit models have higher Brier scores, especially under the Unit Information Prior.

Overall, the results suggest that logistic regression with Gaussian prior performs best among the four fitted models in terms of prediction, while the Unit Information Prior gives more stable effective sample sizes. Therefore, the choice of prior affects different aspects of model performance, and the logistic model with Gaussian prior appears to be the strongest overall in this analysis.

Appendix

1. Values of Yi created:

```
## Yi from Cauchit distribution:
```

```
##   [1] 1 0 0 0 1 0 0 0 0 1 1 0 1 0 1 1 0 0 0 1 0 0 0 0 0 0 0 1 0 1 1 1 0 0 1 0 0
##  [38] 0 1 0 1 1 0 0 1 0 1 1 1 0 1 0 1 1 0 1 0 0 0 0 1 1 1 1 1 1 1 1 0 1 0 0 0 0
##  [75] 1 0 1 0 1 0 1 1 1 0 1 1 1 0 0 1 1 1 0 1 1 1 1 1 1 0 1 0 0 0 1 0 0 0 1 0 1
## [112] 0 0 0 1 0 0 0 0 0 0 1 0 0 0 0 1 0 1 0 0 0 1 0 1 1 1 0 1 1 0 0 1 1 1 0 0 1
## [149] 0 1
```

```
## Yi from Logistic distribution:
```

```
##   [1] 1 0 0 1 1 0 0 0 0 1 1 0 1 0 1 1 0 0 0 1 0 0 0 0 0 0 0 1 1 1 1 1 0 1 1 0 0
##  [38] 0 1 0 1 1 0 0 0 0 1 1 1 0 1 0 0 1 0 1 0 0 0 0 1 1 1 1 1 1 1 1 0 1 0 0 0 0
##  [75] 1 0 1 0 1 0 1 1 1 0 0 1 1 0 0 1 1 1 0 1 1 1 1 1 1 0 1 0 0 0 1 0 0 0 1 0 1
## [112] 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0 1 0 1 1 0 0 1 1 0 0 1 1 1 0 0 1
## [149] 0 1
```

2. Metropolis-Hastings algorithm

```
## First few parameter samples for logistic regression with Gaussian prior:
```

```
##           [,1]      [,2]      [,3]       [,4]      [,5]      [,6]       [,7]
## [1,] -1.904219 -1.083872 -1.646817 -0.8262116 0.6850486 0.2379076  0.2195384
## [2,] -1.904219 -1.083872 -1.646817 -0.8262116 0.6850486 0.2379076  0.2195384
## [3,] -1.985961 -1.038974 -1.606922 -0.9312581 1.9780331 0.2691063 -0.5353209
##          [,8]     [,9]    [,10]
## [1,] 1.188599 1.467704 1.867412
## [2,] 1.188599 1.467704 1.867412
## [3,] 1.203146 1.336366 1.835838
```

```
## First few parameter samples for cauchit regression with Gaussian prior:
```

```
##           [,1]      [,2]      [,3]       [,4]        [,5]      [,6]      [,7]
## [1,] -2.000000 -1.555556 -1.111111 -0.6666667 -0.22222222 0.2222222 0.6666667
## [2,] -1.944490 -1.690102 -1.157401 -0.5821374  0.02111254 0.2444899 0.3068987
## [3,] -2.128366 -1.534635 -1.275968 -0.6778127 -0.51020135 0.2559266 0.1543599
##          [,8]     [,9]    [,10]
## [1,] 1.111111 1.555556 2.000000
## [2,] 1.173580 1.524004 1.866352
## [3,] 1.124195 1.570438 1.865302
```

```
## Acceptance rate (logistic regression with Gaussian prior): 0.45706
```

```
## Acceptance rate (cauchit regression with Gaussian prior): 0.45602
```

```
## First few parameter samples for logistic regression with Unit Information Prior:
```

```
##           [,1]      [,2]      [,3]       [,4]       [,5]      [,6]      [,7]
## [1,] -2.000000 -1.555556 -1.111111 -0.6666667 -0.2222222 0.2222222 0.6666667
## [2,] -1.897127 -1.466287 -1.212422 -0.3093527 -0.7104465 0.2173968 0.6853385
## [3,] -1.897127 -1.466287 -1.212422 -0.3093527 -0.7104465 0.2173968 0.6853385
##          [,8]     [,9]  [,10]
## [1,] 1.111111 1.555556 2.0000
## [2,] 1.129105 1.602992 2.0728
## [3,] 1.129105 1.602992 2.0728
```

```
## First few parameter samples for cauchit regression with Unit Information Prior:
```

```
##           [,1]      [,2]       [,3]       [,4]       [,5]      [,6]     [,7]
## [1,] -1.914096 -1.844604 -0.7733841 -0.6942183 -0.4431989 0.1618249 1.923107
## [2,] -1.995021 -2.052355 -0.5785837 -0.7167103  0.4758883 0.1717194 1.858979
## [3,] -1.925944 -2.025517 -0.5592641 -0.8745294 -0.4505223 0.1281197 2.661677
##          [,8]     [,9]    [,10]
## [1,] 1.145898 1.564111 2.046915
## [2,] 1.333360 1.457663 1.920156
## [3,] 1.295678 1.549731 2.096485
```

```
## Acceptance rate (logistic regression with Unit Information Prior): 0.73444
```

```
## Acceptance rate (cauchit regression with Unit Information Prior): 0.72438
```